



Heaven's Light is Our Guide

Department of Computer Science & Engineering
Rajshahi University of Engineering & Technology, Bangladesh

Course Code: CSE 2203
Course Title: Digital Techniques

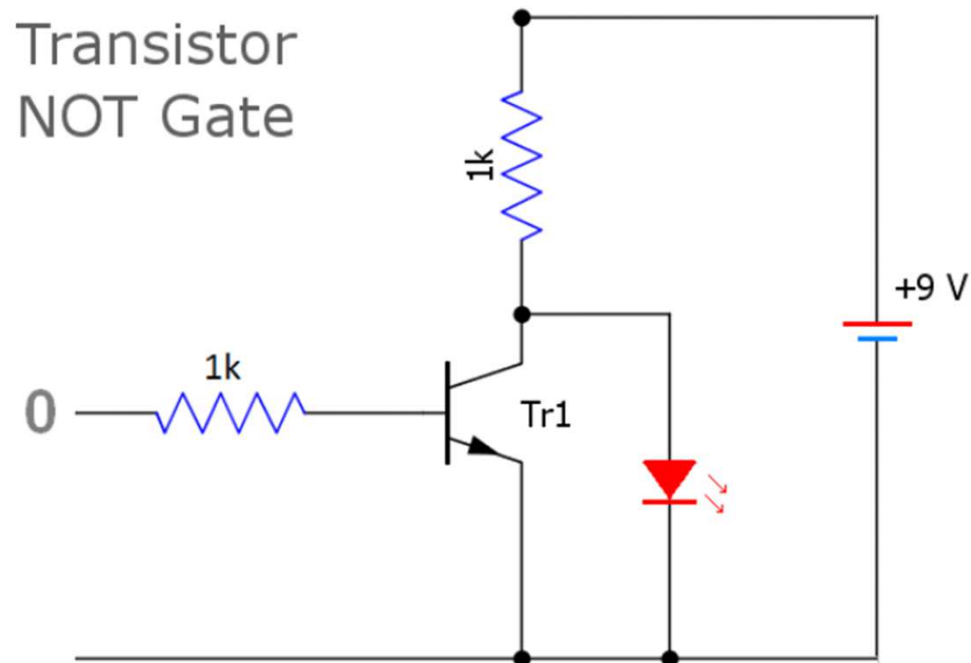
Presented by,
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The TTL Logic Family

- Most TTL circuits have a similar structure
 - **NAND** and **AND** gates use multiple-emitter transistor or multiple diode junction inputs.
 - **NOR** and **OR** gates use separate input transistors.
- The input will be the cathode of a P-N junction
 - A HIGH input will turn off the junction.
 - Only a leakage current is generated.
 - A LOW input turns on the junction.
 - Relatively large current is generated.
- Most TTL circuits have some type of totem-pole output configuration.

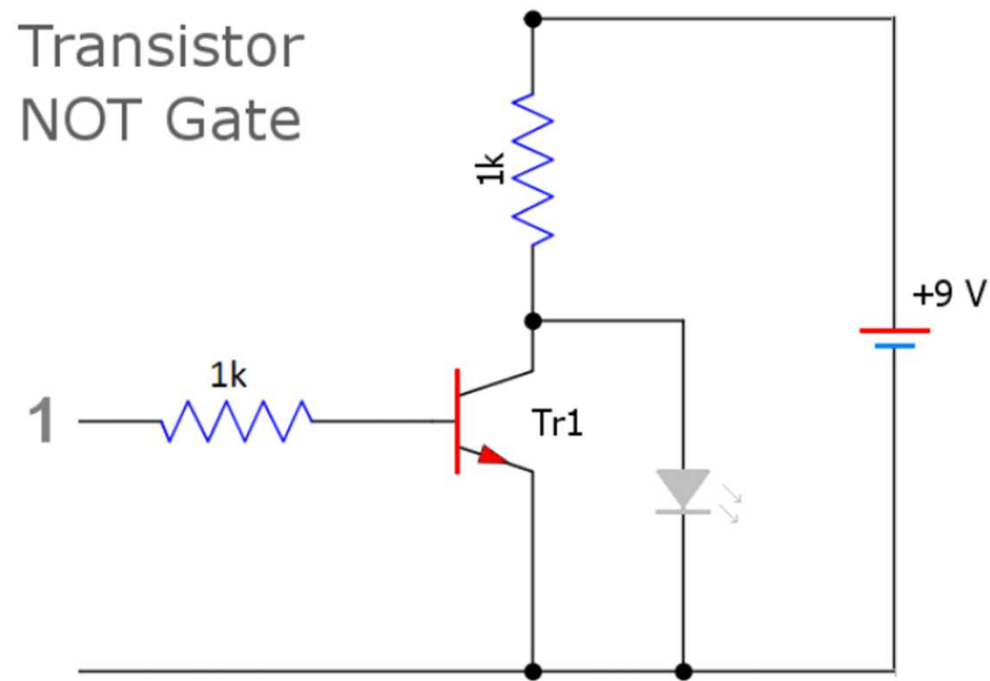
The TTL Logic Family

1. Internal circuit for a TTL NOT gate (Case 1)



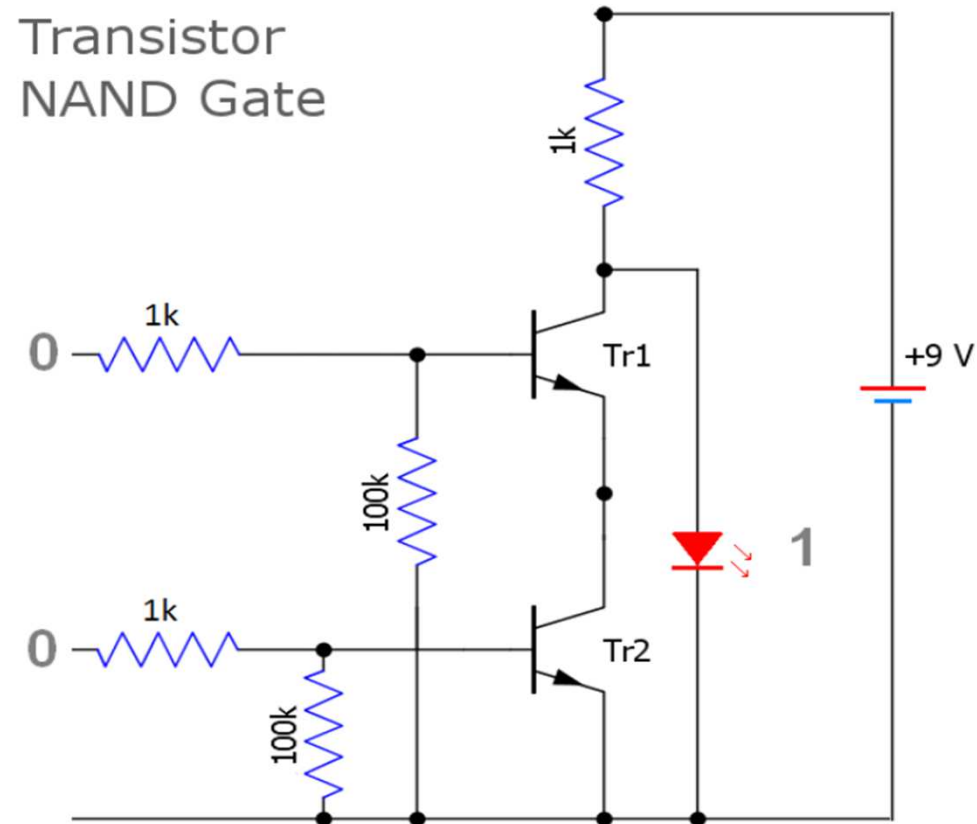
The TTL Logic Family

1. Internal circuit for a TTL NOT gate (Case 2)



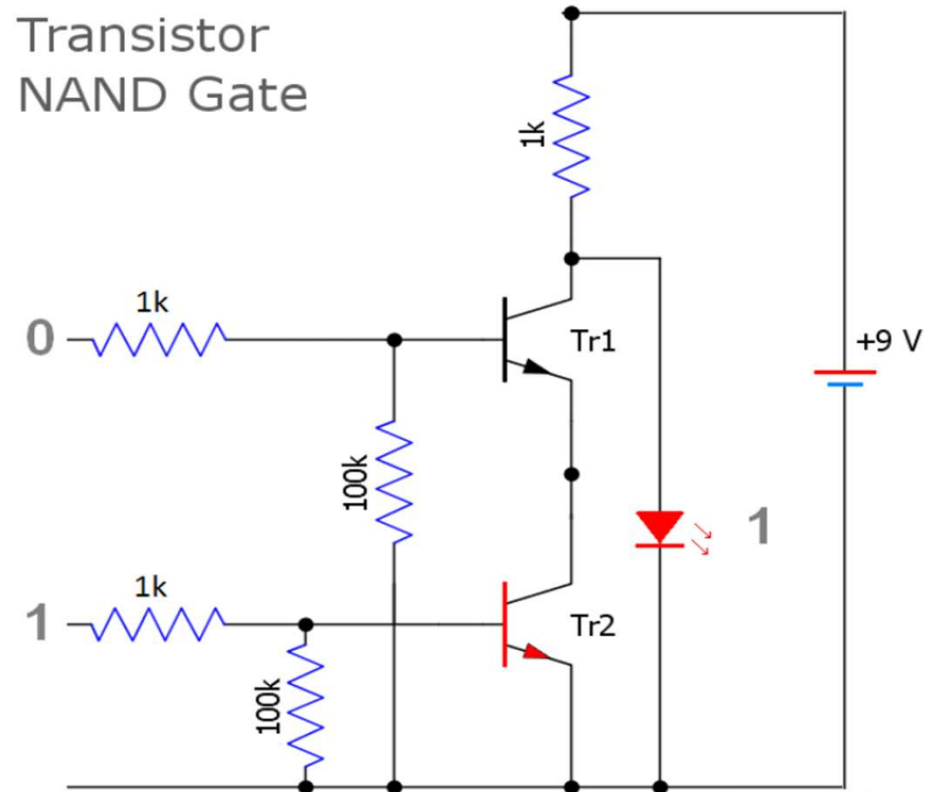
The TTL Logic Family

2. Internal circuit for a TTL NAND gate (Case 1)



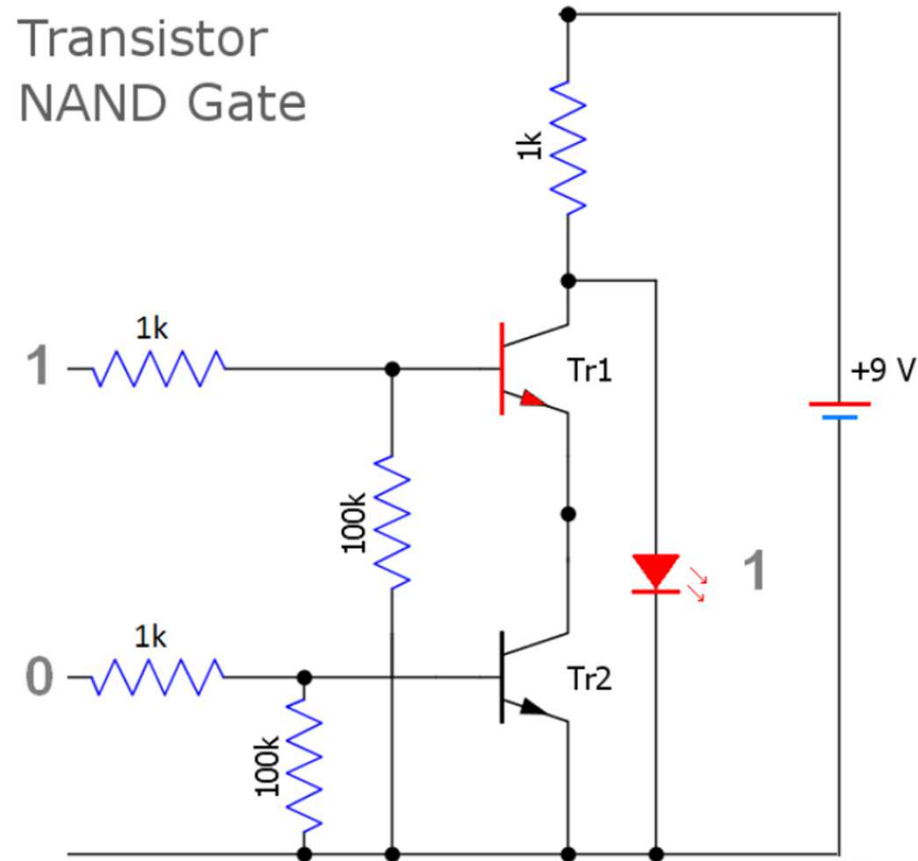
The TTL Logic Family

2. Internal circuit for a TTL NAND gate (Case 2)



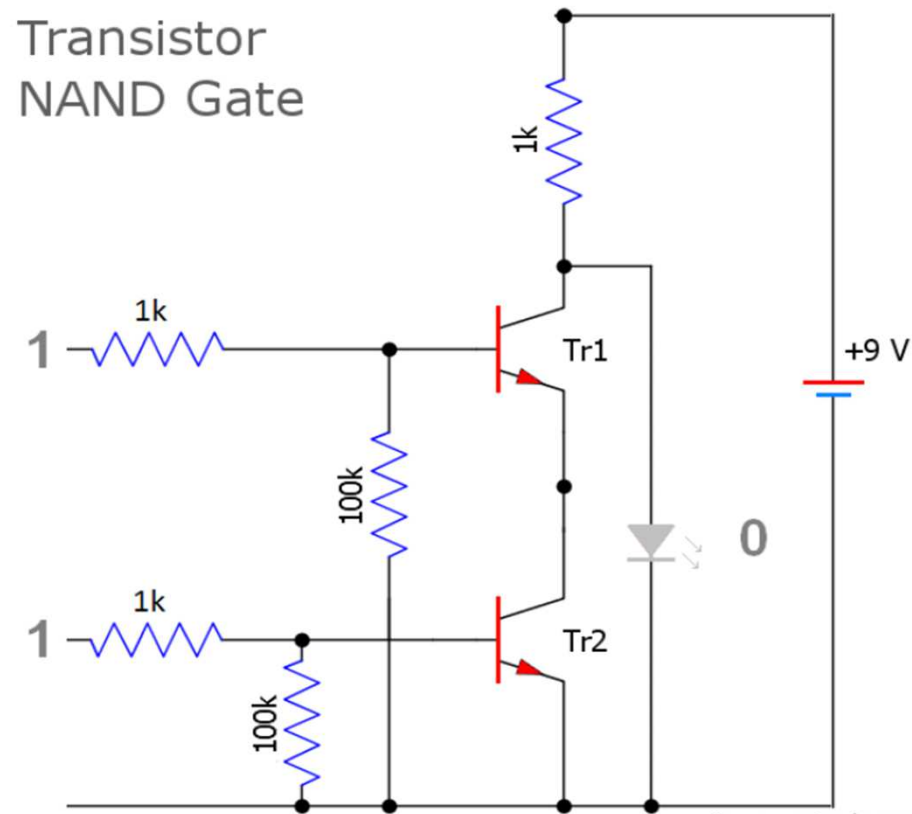
The TTL Logic Family

2. Internal circuit for a TTL NAND gate (Case 3)



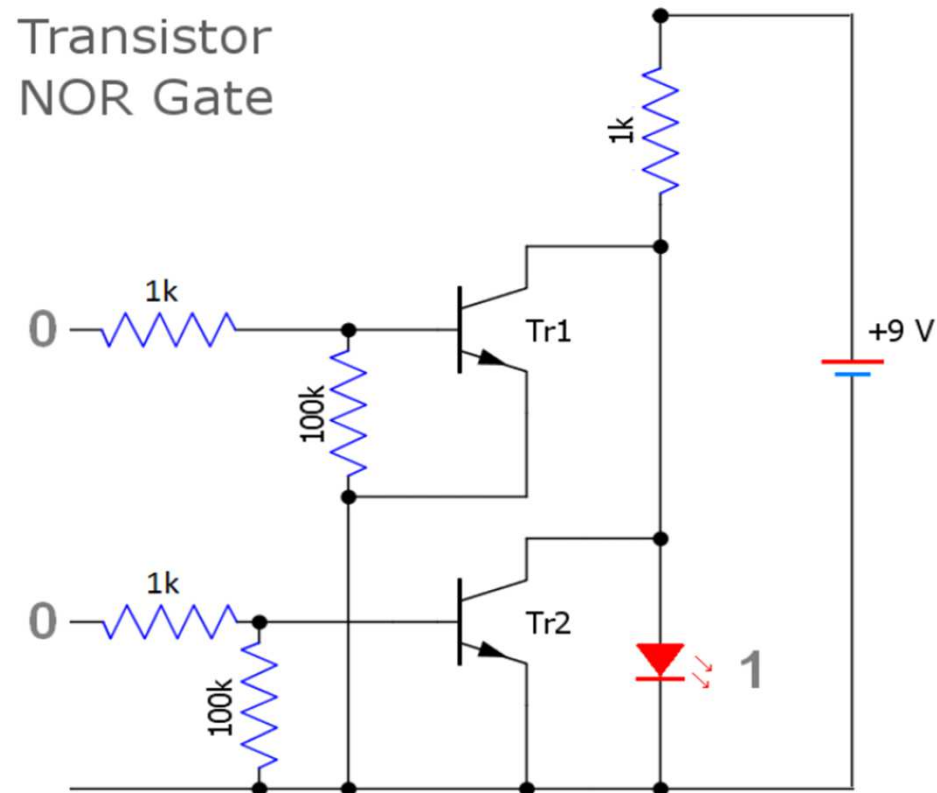
The TTL Logic Family

2. Internal circuit for a TTL NAND gate (Case 4)



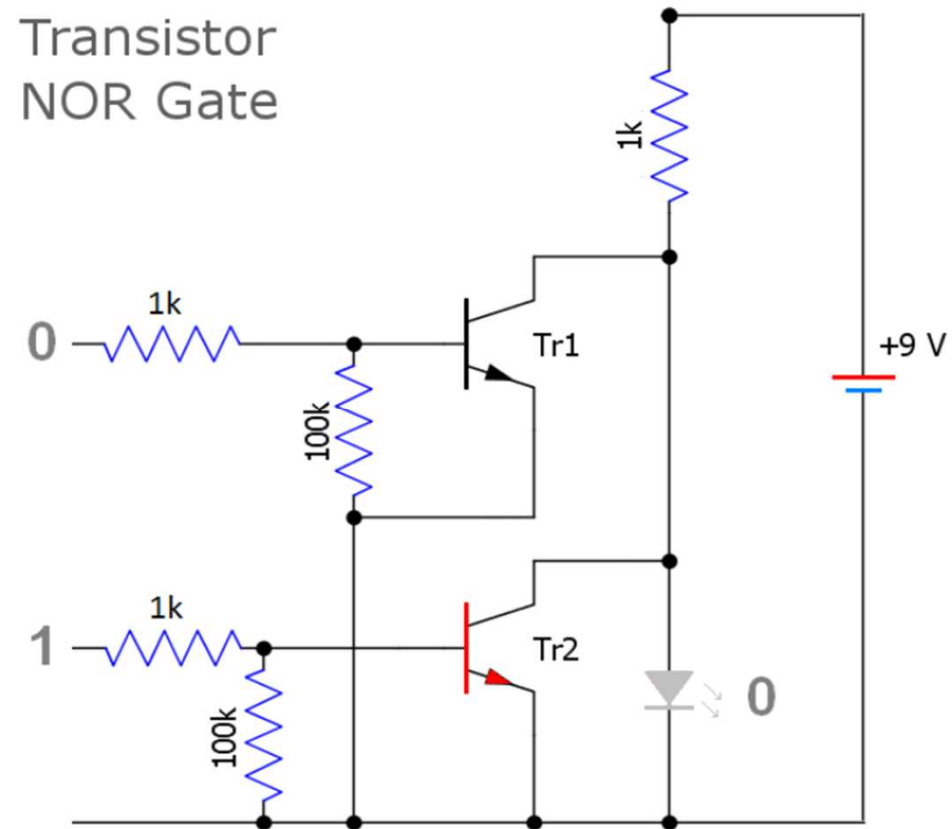
The TTL Logic Family

3. Internal circuit for a TTL NOR gate (Case 1)



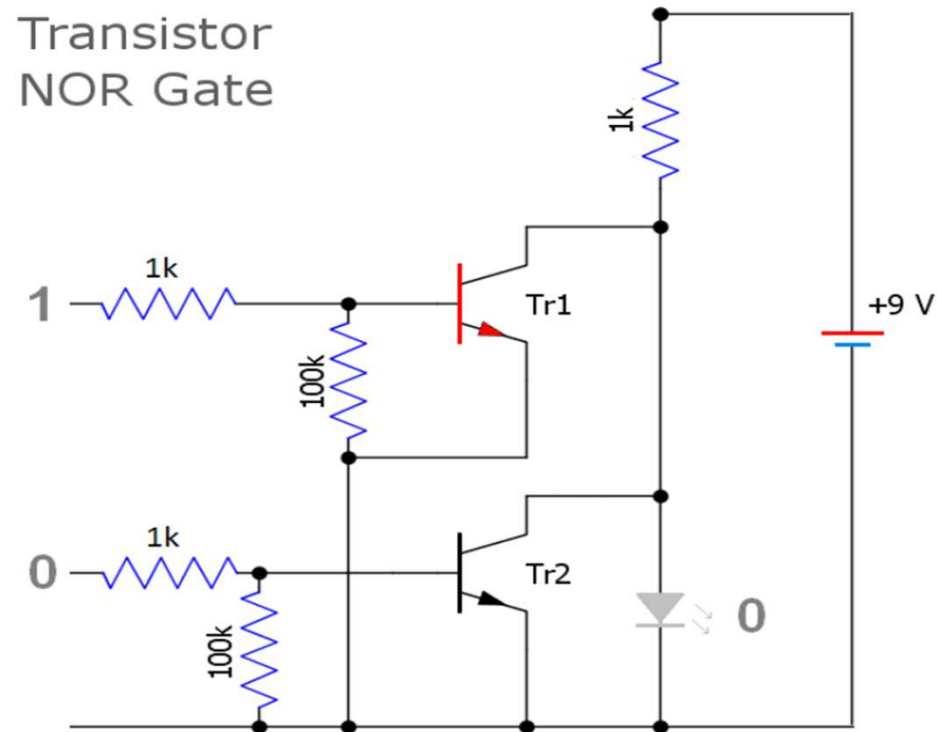
The TTL Logic Family

3. Internal circuit for a TTL NOR gate (Case 2)



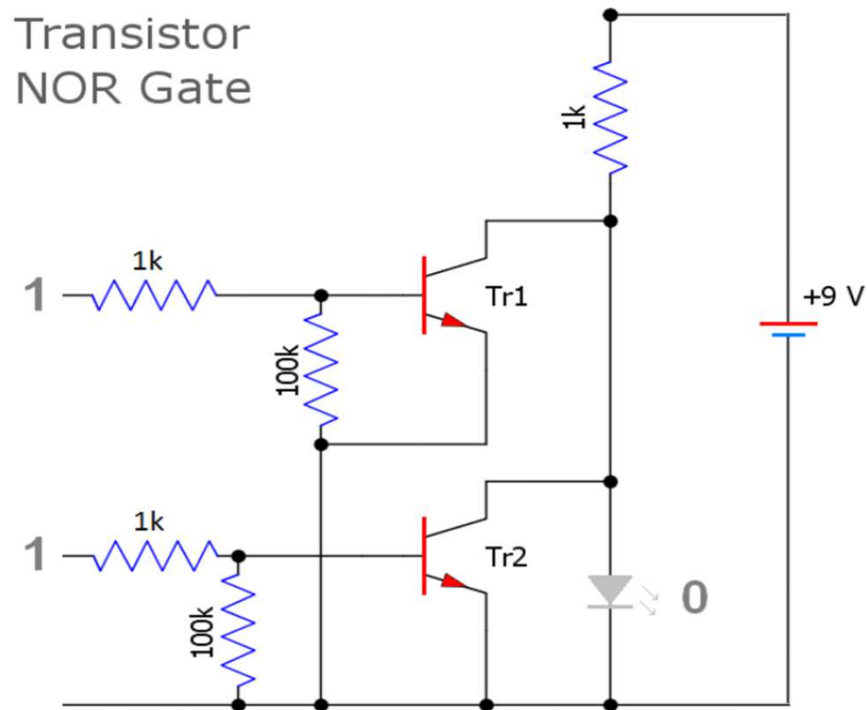
The TTL Logic Family

3. Internal circuit for a TTL NOR gate (Case 3)



The TTL Logic Family

3. Internal circuit for a TTL NOR gate (Case 4)



Some Questions/Home Works

1. How can you implement **AND** gate using **TTL** Logic?
2. How can you implement **OR** gate using **TTL** Logic?
3. How can you implement **Boolean function** using **TTL** Logic ?

MOS Technology

- MOS technology derives its name from the basic structure of a metal electrode, over an oxide insulator, over a semi-conductor substrate.
 - Transistors of MOS technology are field-effect transistors—called MOSFETs.

The electric *field* on the *metal* electrode side of the *oxide* insulator has an *effect* on the resistance of the substrate.

MOS Technology

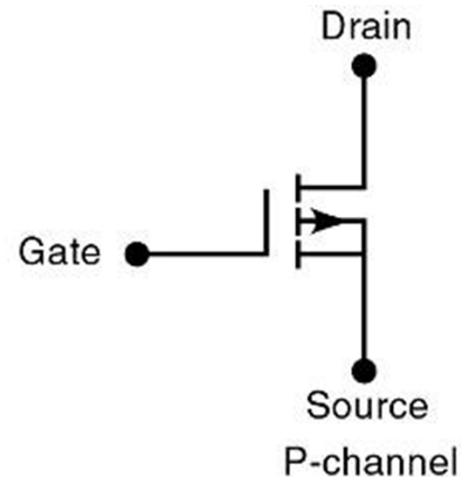
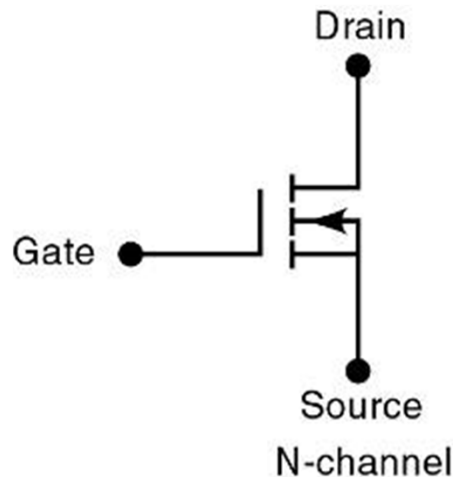
- Most of the MOS digital ICs are constructed entirely of MOSFETs and no other components.

Advantages of MOSFETS

- MOSFETs are
 1. relatively simple
 2. inexpensive to fabricate
 3. small and
 4. consume very little power

MOS Technology

- There are presently two general types of MOSFETs—*depletion* and *enhancement*.
 - MOS ICs use enhancement MOSFETs exclusively.



The direction of the arrow indicates either P- or N-channel. The symbols show a broken line between the *source* and the *drain* to indicate there is *normally* no conducting channel between these electrodes.

MOS Technology – Basic MOSFET Switch

- An N-channel MOSFET is the basic element in a family of devices known as **N-MOS**.
- Drain is always biased positive relative to the source.

MOS Technology – Basic MOSFET Switch

- Gate-to-source voltage V_{GS} is the input voltage.
- Where, $V_{GS} = V_G - V_S$
- ON/OFF Condition of N-MOS:
 - a) When $V_{GS} \geq V_{th}$ then NMOS is ON*
 - b) When $V_{GS} < V_{th}$ then NMOS is OFF*

MOS Technology – Basic MOSFET Switch

- The P-channel MOSFET—**P-MOS**—operates in the same manner as the N-channel.
 - Except that it uses voltages of opposite polarity.
- The **drain is connected to the lower side** of the circuit so it is biased with a more negative voltage relative to the source.

MOS Technology – Basic MOSFET Switch

- Gate-to-source voltage V_{GS} is the input voltage.
- Where, $V_{SG} = V_S - V_G$
- ON/OFF Condition of P-MOS:
 - a) When $V_{SG} \geq V_{th}$ then PMOS is ON*
 - b) When $V_{SG} < V_{th}$ then PMOS is OFF*

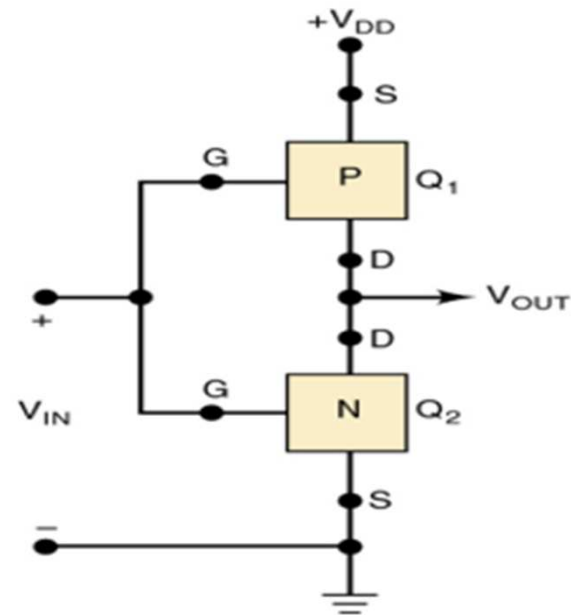
Complementary MOS Logic – CMOS Inverter

- P-MOS & N-MOS circuits began to dominate the LSI and VLSI markets in the 1970s and 1980s.
 - Use fewer components & are much simpler to manufacture than TTL circuits.
- During this era, technology emerged that used P-MOS & N-MOS transistors in the same circuit.
 - Complementary MOS, or **CMOS**, technology.

CMOS Inverter

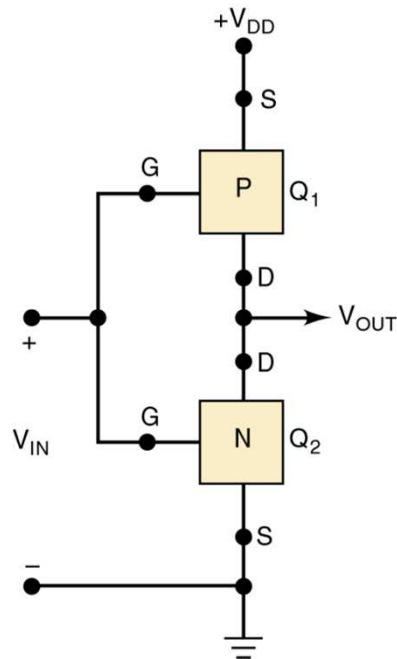
1. Implementing NOT gate

- Not Gate: $f = \bar{A}$
- Pulldown: $\bar{f} = A$
- Pullup: $f = \bar{A}$



Complementary MOS Logic – CMOS Inverter

- The CMOS INVERTER has two MOSFETs in series.
 - The P-channel device source is connected to V_{DD} .
 - The N-channel device has its source connected to ground—usually labeled V_{SS} .



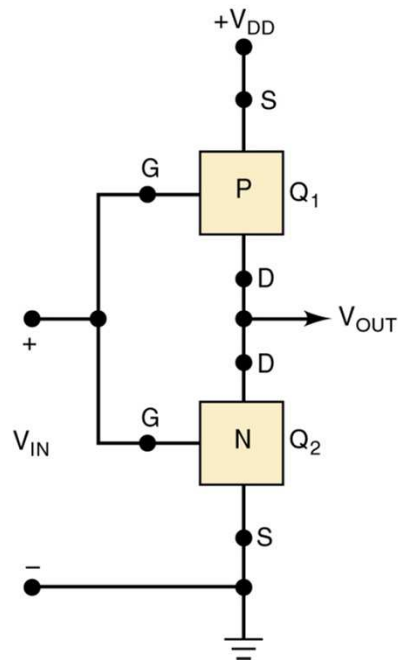
V_{IN}	Q_1	Q_2	V_{OUT}
$+V_{DD}$ (logic 1)	OFF $R_{OFF} = 10^{10} \Omega$	ON $R_{ON} = 1 \text{ k}\Omega$	$\approx 0 \text{ V}$
0 V (logic 0)	ON $R_{ON} = 1 \text{ k}\Omega$	OFF $R_{OFF} = 10^{10} \Omega$	$\approx +V_{DD}$

$$V_{OUT} = \overline{V_{IN}}$$

Basic CMOS INVERTER.

Complementary MOS Logic – CMOS Inverter

- The CMOS INVERTER has two MOSFETs in series.
 - Gates of the two devices are connected together as a common input.
 - Drains are connected together as common output.



V_{IN}	Q_1	Q_2	V_{OUT}
$+V_{DD}$ (logic 1)	OFF $R_{OFF} = 10^{10} \Omega$	ON $R_{ON} = 1 \text{ k}\Omega$	$\approx 0 \text{ V}$
0 V (logic 0)	ON $R_{ON} = 1 \text{ k}\Omega$	OFF $R_{OFF} = 10^{10} \Omega$	$\approx +V_{DD}$

$$V_{OUT} = \overline{V_{IN}}$$

Basic CMOS INVERTER.

Complementary MOS Logic

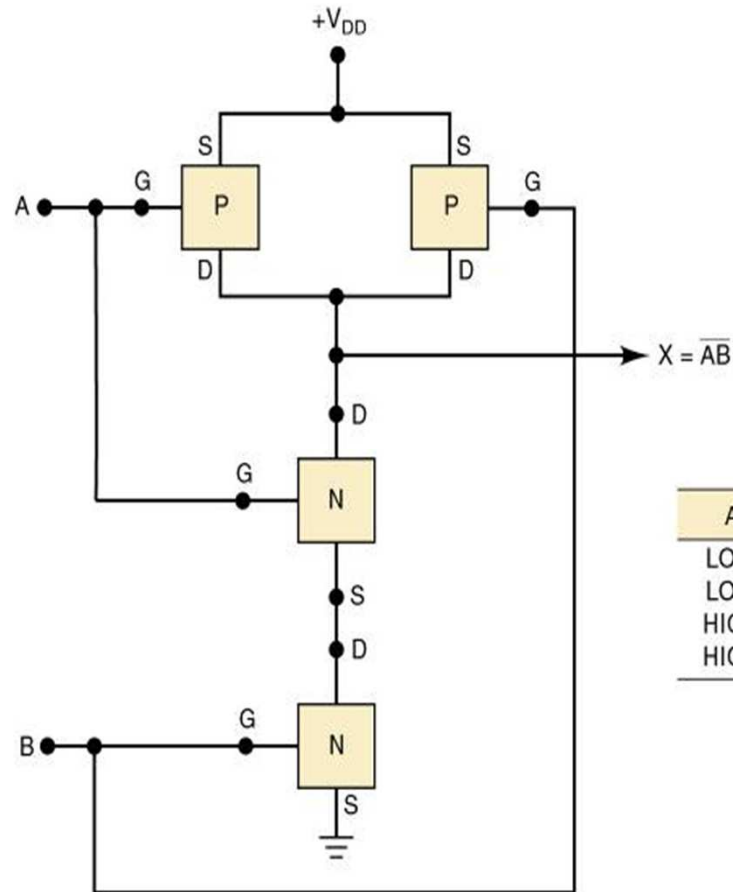
2. Implementing **NAND** gate

For NAND gate, $f = \overline{A.B}$

Pulldown $\bar{f} = A.B$

Pullup $p = f = \overline{A.B}$
 $= \bar{A} + \bar{B}$

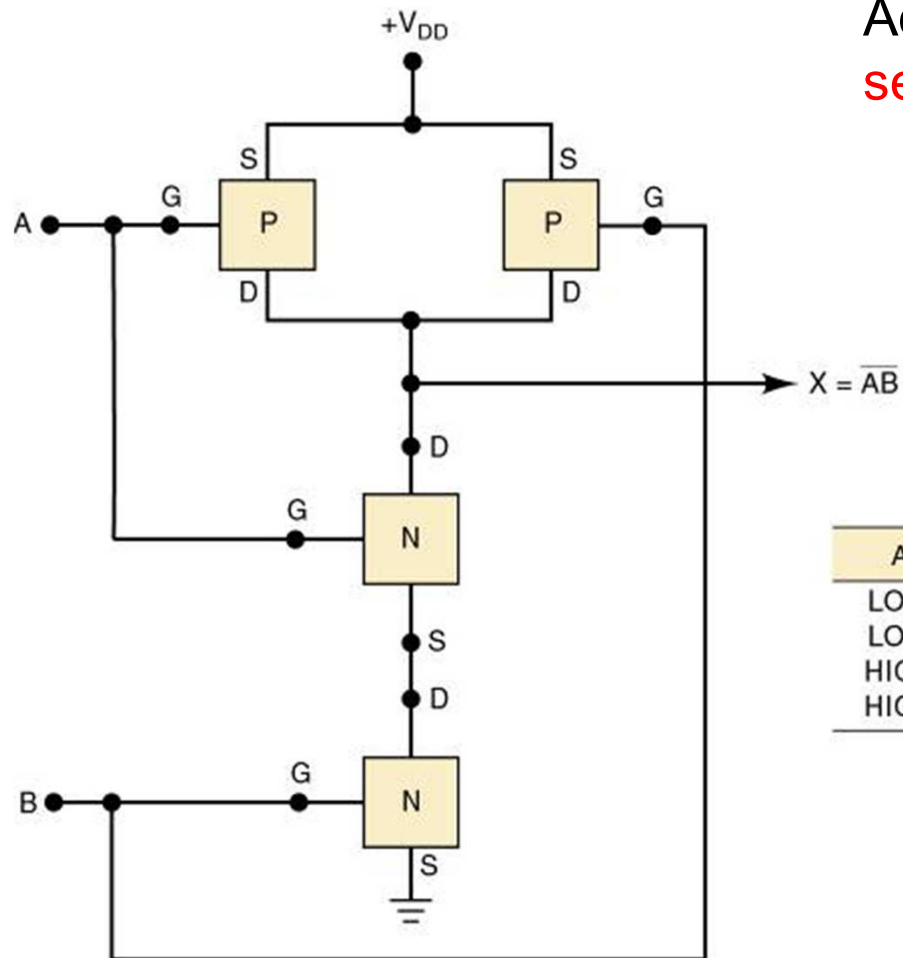
(De Morgan's Laws)



A	B	X
LOW	LOW	HIGH
LOW	HIGH	HIGH
HIGH	LOW	HIGH
HIGH	HIGH	LOW

Complementary MOS Logic

- A **NAND** gate is formed by modifying the basic INVERTER.



Adding **parallel P-channel & series N-channel** MOSFETs to the basic INVERTER.

A	B	X
LOW	LOW	HIGH
LOW	HIGH	HIGH
HIGH	LOW	HIGH
HIGH	HIGH	LOW

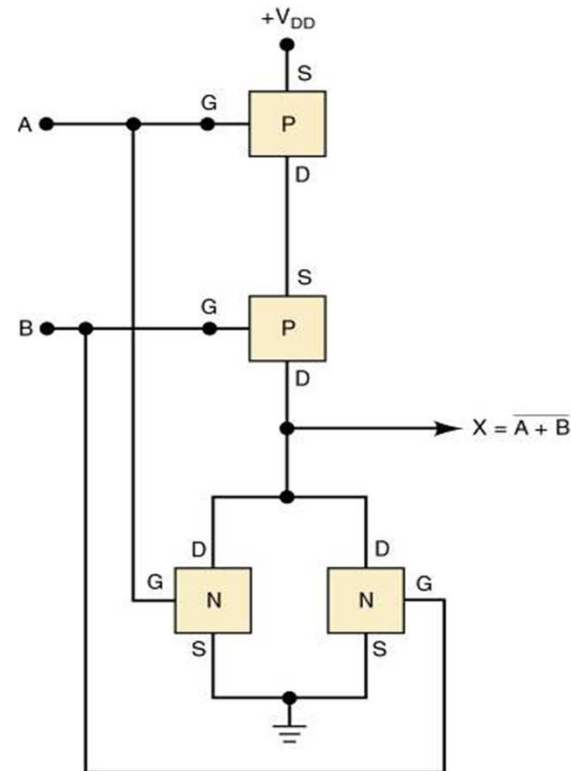
Complementary MOS Logic

3. Implementing **NOR** gate

For NOR gate, $f = \overline{A+B}$

Pulldown $\bar{f} = A+B$

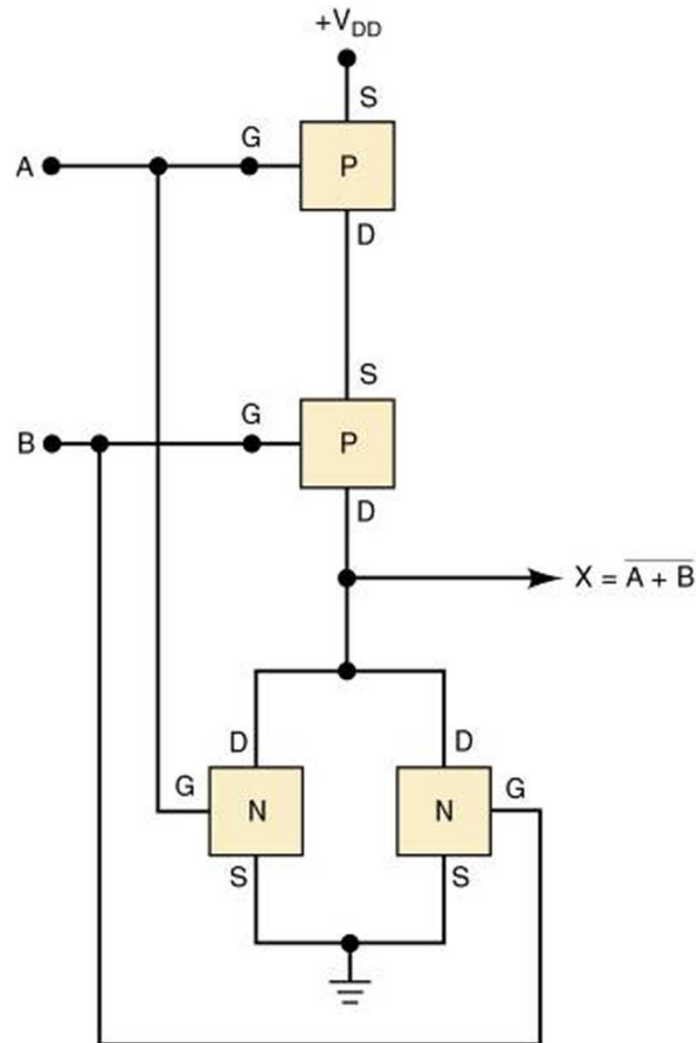
Pullup $p = f = \overline{A+B}$
 $= \bar{A} \cdot \bar{B}$



A	B	X
LOW	LOW	HIGH
LOW	HIGH	LOW
HIGH	LOW	LOW
HIGH	HIGH	LOW

Complementary MOS Logic

- A CMOS **NOR** gate.



Formed by adding a **series P-MOS** and a **parallel N-MOS** to the basic INVERTER.

A	B	X
LOW	LOW	HIGH
LOW	HIGH	LOW
HIGH	LOW	LOW
HIGH	HIGH	LOW

Some Questions/Home Works

1. How can you implement **AND** gate using CMOS Logic?
2. How can you implement **OR** gate using CMOS Logic?
3. How can you implement **Boolean function** using CMOS (**$f = A + BC$**)?

**EVERY ENDING
IS REALLY JUST A
NEW BEGINNING**

Thank You